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Empirical Study on Accuracy of LLM-Generated Choropleth Maps vs Programmatic Visualization

1. Project Overview

Choropleth maps are widely used for representing region-wise data such as population, student distribution, or resource allocation. With the advent of Large Language Models (LLMs) like ChatGPT, it is now possible to generate such maps using natural language prompts.

However, the correctness of such AI-generated outputs is uncertain, especially in spatial representations.

This project aims to conduct an **empirical comparison** between:

1. Choropleth maps generated using LLM prompts
2. Choropleth maps generated using structured Python-based methods

The study focuses on identifying:

- accuracy differences
- sources of errors
- reliability of LLM outputs

2. Objectives

1. To understand choropleth map visualization concepts
2. To generate maps using both LLM and programmatic approaches
3. To compare outputs for correctness and consistency
4. To identify sources of error in LLM-generated outputs
5. To develop analytical and reporting skills

3. Problem Statement

Given a **district-wise dataset** (e.g., **Karnataka districts**), generate choropleth maps using:

- Approach 1: LLM-generated code and output
- Approach 2: Manual Python implementation

Compare the results and evaluate:

Which approach produces a more accurate representation and why?

4. Technologies to be Used

Component	Tool
LLM Interaction	ChatGPT
Programming	Python
Visualization	Matplotlib / basic mapping tools
Data Format	CSV

5. Methodology

Phase 1: LLM-Based Map Generation

- Prompt LLM to:
 - generate Python code
 - generate choropleth map
- Execute and capture output

Phase 2: Manual Map Generation

- Use Python to:
 - load dataset
 - map district names
 - generate choropleth map correctly

Phase 3: Comparison & Evaluation

Compare both outputs based on:

- correctness of district mapping
- consistency of values
- visual accuracy
- completeness

Phase 4: Error Analysis

Identify reasons for mismatch such as:

- incorrect district names
- missing regions
- improper data mapping
- lack of spatial grounding in LLM

Phase 5: Reporting

Students prepare a report including:

- methodology
- observations
- comparison tables
- conclusions

6. Expected Outputs

- LLM-generated choropleth map
- Python-generated correct choropleth map
- Comparison table
- Final report (analysis + conclusion)

7. Learning Outcomes

After completing the project, students will:

- understand structured vs unstructured data processing
- learn limitations of LLMs in spatial tasks
- gain basic data visualization skills
- develop analytical thinking and evaluation ability